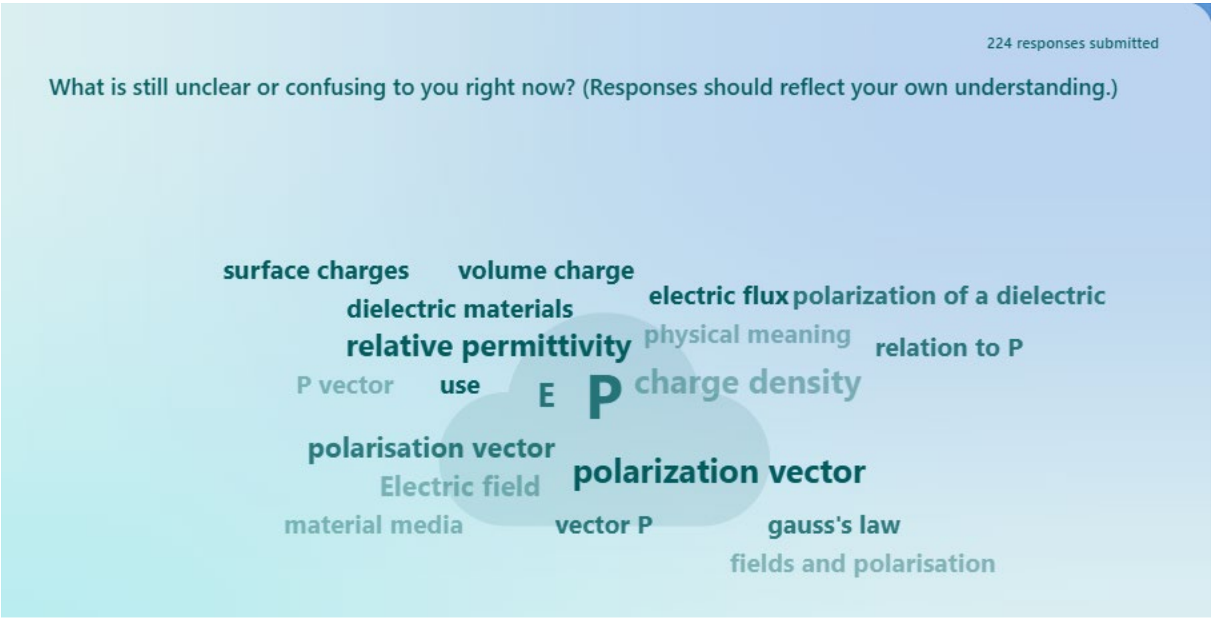
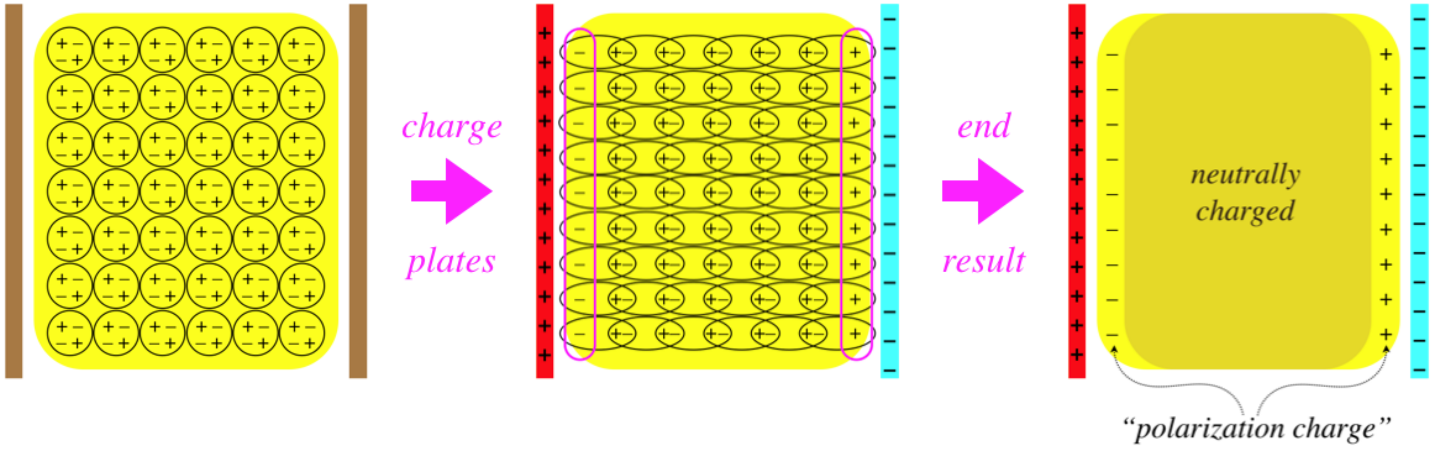
ELEC 3115

Lecture 3: Week 2 , Monday

Your responses at the end of lecture 2:



What Exactly Is the Polarization Vector?



$$\rho_p = -\vec{\nabla} \bullet \vec{P}$$

First postulate in dielectric: $\vec{\nabla} \bullet \vec{E} = \frac{1}{\varepsilon_0}\left(\rho + \rho_p\right)$

Substituting, $\vec{\nabla} \bullet \varepsilon_0 \vec{E} = \left(\rho - \vec{\nabla} \bullet \vec{P}\right)$

Rearranging, $\vec{\nabla} \bullet \left(\varepsilon_0 \vec{E} + \vec{P}\right) = \rho$

Comparing with $\vec{\nabla} \bullet D = \rho$

We found,

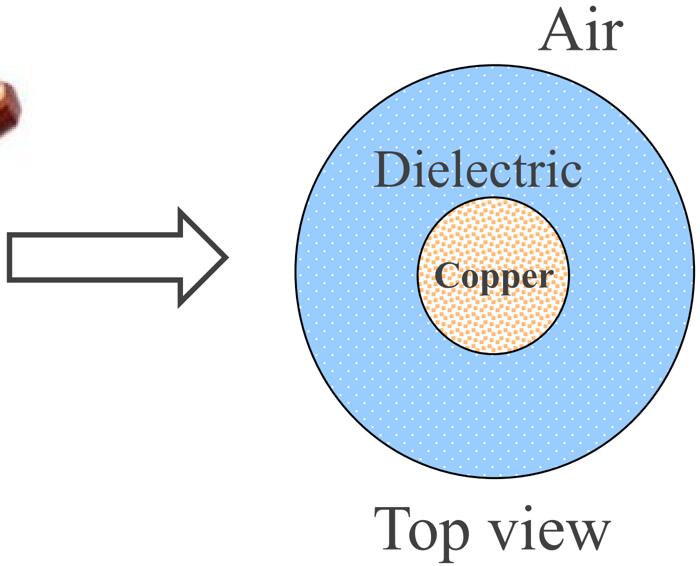
$$\varepsilon_0 \vec{E} + \vec{P} = \overrightarrow{D} \rightarrow \vec{P} = \overrightarrow{D} - \varepsilon_0 \vec{E} = \overrightarrow{D} - \overrightarrow{D}_{\text{freespace}}$$

- Polarization vector **P** = Dipole moments in a unit volume, difficult to measure in practice as dipoles are at the microscopic level.

- Polarization increases with applied **E**. $\vec{P} \propto \vec{E} \rightarrow \vec{P} = \chi_e \varepsilon_0 \vec{E}$

$$\vec{\nabla} \bullet \left(\varepsilon_0 \vec{E} + \vec{P}\right) = \rho \rightarrow \vec{\nabla} \bullet \left(\varepsilon_0 \vec{E} + \chi_e \varepsilon_0 \vec{E}\right) = \rho \rightarrow \vec{\nabla} \bullet \left(1 + \chi_e\right) \varepsilon_0 \vec{E} = \rho \rightarrow \vec{\nabla} \bullet \varepsilon_r \varepsilon_0 \vec{E} = \rho$$

- Relative permittivity ε_r is a factor by which the electric field **E** is reduced in a dielectric material relative to vacuum/free space.



- How does the electric field of the copper conductor change at the insulating jacket? What happens at the boundary of the two materials?
- What factors influence the design of the cable's insulating jacket?
- How capacitance forms around charged bodies?

Today's learning goals

By the end of today's lecture, you will be able to:

- Analyse the behaviour of the electric field at the boundary of two different materials using *boundary conditions*.
- Design a cable by selecting appropriate dielectric materials and their thickness based on voltage specifications.
- Explain capacitance formations in transmission lines and cables, and calculate them from $\mathbf{E}$ and V.

Electric Field at Material Boundaries

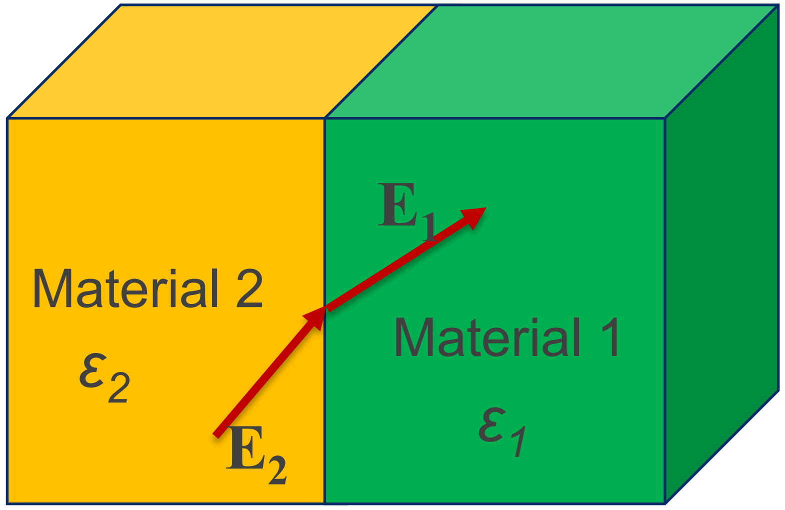
Why this matters:
Real engineering systems are made of multiple materials.
Electric fields behave differently at interfaces.

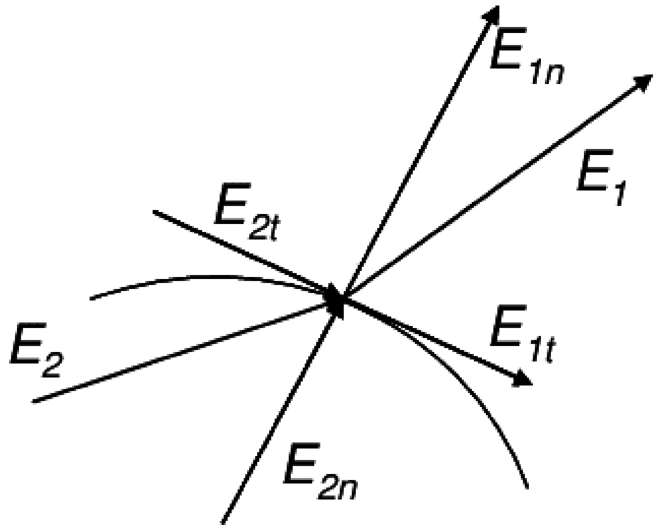
In this section, you will:

- Analyse the normal and tangential components of $\mathbf{E}$
- Apply boundary conditions at conductor–dielectric interfaces
- Understand how permittivity changes the electric field strength

Electric field at boundary (interface) of two media

1. Electric field at the boundary of a conductor and free space (and dielectric).
2. Electric field at the boundary of two different dielectrics.



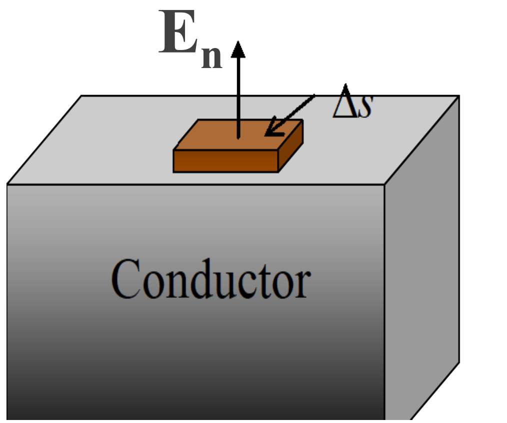


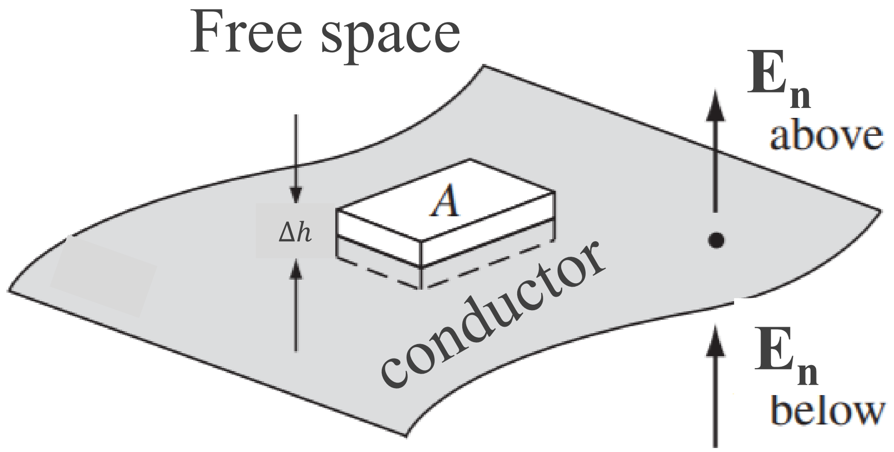
$\mathbf{E}_n$: Normal component, which is normal on the boundary surface.
$\mathbf{E}_t$: Tangential component, which is tangential or parallel on the boundary surface.

Boundary conditions at the interface of a conductor and free space

E inside the conductor is zero, but surface charge density ρ_s exists.

To check $\mathbf{E_n}$ at the boundary, consider a very thin pillbox of surface area of *Δs (shown as A in the right-hand figure)* at the interface with negligible height (*Δh→0). Δs* is the Gaussian surface area for $\mathbf{E_n}$.





From Gauss's Law (postulate 1), $\oint_s \vec{E}.d\vec{s} = \frac{Q_{enclosed}}{\varepsilon_0} \rightarrow E_n \Delta s = \frac{\rho_s \Delta s}{\varepsilon_0}$

$|E_n| = \frac{\rho_s}{\varepsilon_o}$, Also $|D_n| = \rho_s$

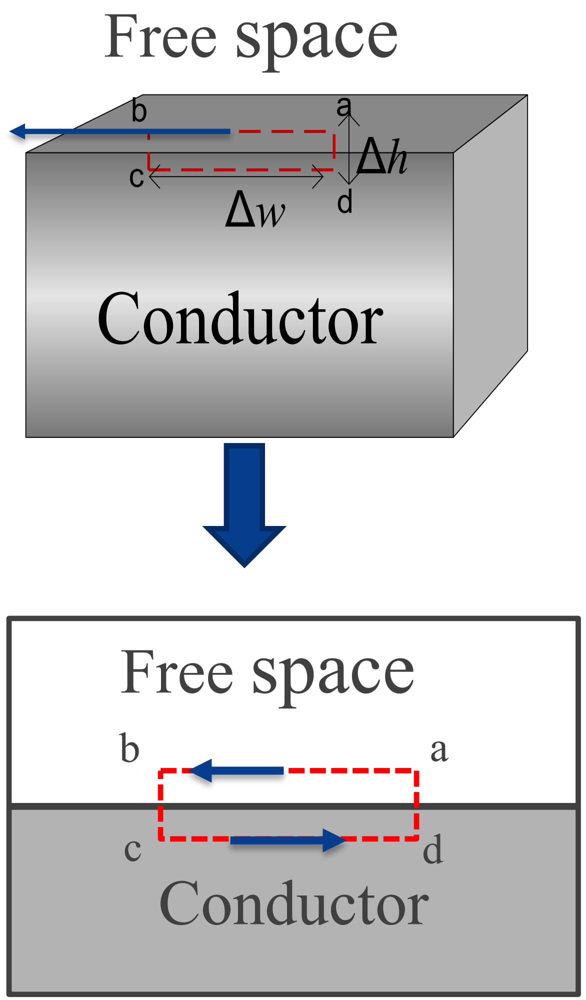
What is $\mathbf{E_n}$ below the surface?

To check $\mathbf{E}_t$, consider a closed contour abcd of height → 0 and width Δw at the boundary.

From Postulate 2,

$$\oint_{abcd} \vec{E} \bullet d\vec{l} = 0$$

$$\int_a^b \vec{E}_{ab} \bullet d\vec{l} + \int_b^c \vec{E}_{bc} \bullet d\vec{l} + \int_c^d \vec{E}_{cd} \bullet d\vec{l} + \int_d^a \vec{E}_{da} \bullet d\vec{l} = 0$$



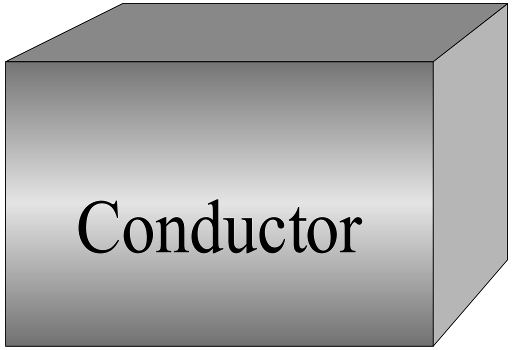
$\Delta h \rightarrow 0$, i.e. length of bc and ad.
ab and cd equal to Δw. $\mathbf{E}_{cd}$ is inside the conductor, hence zero.

$$|E_t \Delta w| = 0$$

$$\therefore |E_t| = 0$$

Also, $|D_t| = 0$

What Just Happened Physically in the conductor?



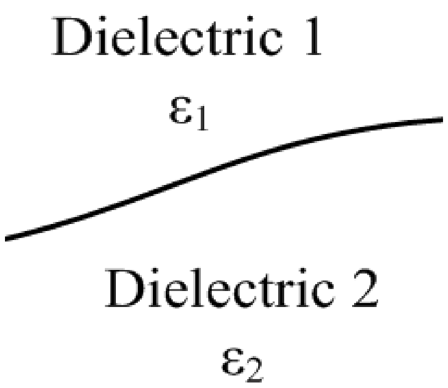
- Inside conductor: $E = 0$
- Just outside: $E_n = \rho_s/\varepsilon_0$
- Tangential component $E_t = 0$

Meaning:

Field lines leave the conductor perpendicular to the surface.

Boundary conditions at the interface of two dielectrics

Interface may or may not have a surface charge density ρ_s.



Tangential components

$$\oint_{abcd} \vec{E} \bullet d\vec{l} = 0$$

$$\left| E_{1t}\Delta w - E_{2t}\Delta w \right| = 0$$

$$\therefore \left| E_{1t} \right| = \left| E_{2t} \right| \quad \text{V/m}$$

$$\because \vec{D}_t = \varepsilon \vec{E}_t \rightarrow \left| \frac{D_{1t}}{\varepsilon_1} \right| = \left| \frac{D_{2t}}{\varepsilon_2} \right| \quad \text{C/m}^2$$

Normal components

$$\oint \vec{D} \bullet d\vec{s} = Q$$

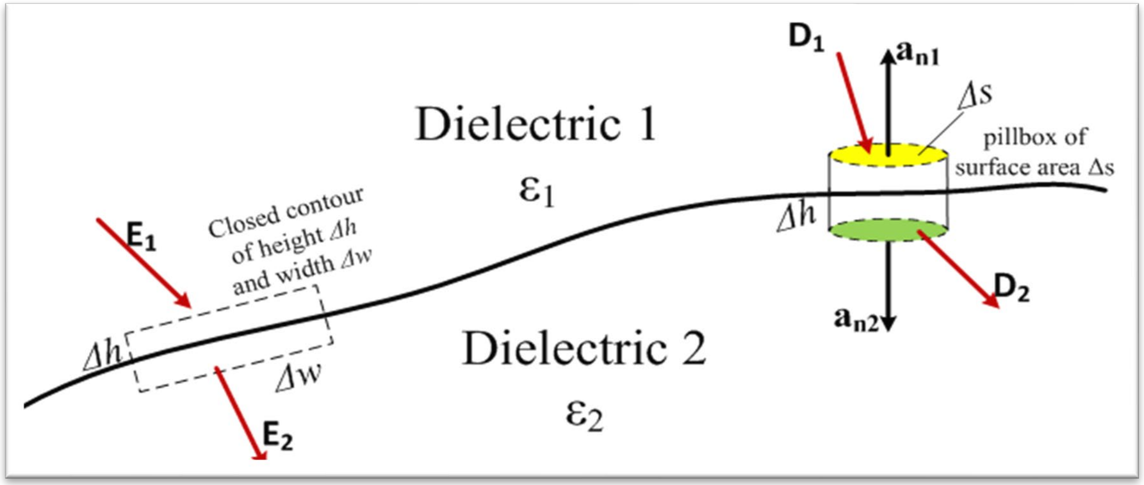
$$\left(-\left| D_{1n} \right| \vec{a}_{n1} \bullet \left| \Delta s \right| \vec{a}_{n1} + \left| D_{2n} \right| \vec{a}_{n2} \bullet \left| \Delta s \right| \vec{a}_{n2} \right) = \rho_s \Delta s$$

note, $\vec{a}_{n2} = -\vec{a}_{n1}$

$$\therefore \left(-D_{1n} + D_{2n} \right) = \rho_s$$

If ρ_s is zero, $\left| D_{1n} \right| = \left| D_{2n} \right|$ or $\varepsilon_1 \left| E_{1n} \right| = \varepsilon_2 \left| E_{2n} \right|$

For dielectric-dielectric interface:

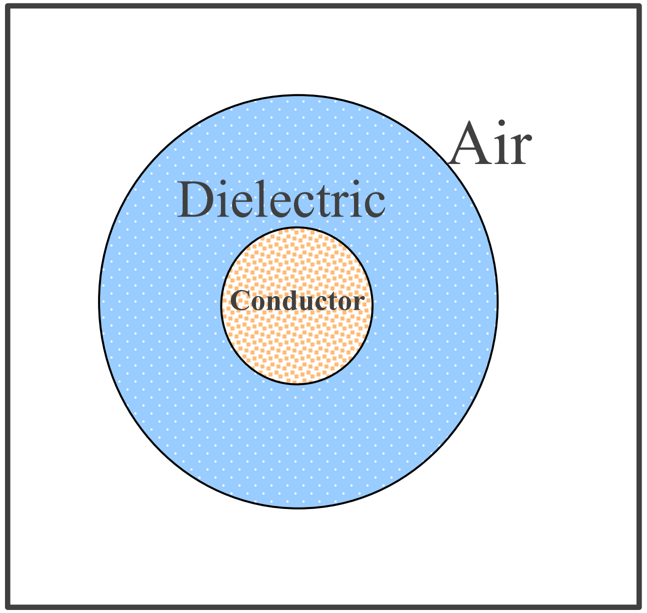


If No Surface Charge Exists:

- Normal $\mathbf{D}_n$ is continuous.
- Tangential $\mathbf{E}_t$ is continuous
- Normal $\mathbf{E}_n$ changes according to permittivity

Meaning:
Higher ε → smaller electric field.

Cable Boundary Summary



Interface	$\mathbf{E_n}$	$\mathbf{E_t}$
Conductor-Dielectric	ρ_s/ε	0
Dielectric-Dielectric (air)	Depends on ε	Continuous

Which interface is more prone to breakdown?

Do we have $\mathbf{E_t}$ in this system at the air-dielectric boundary?

⚠ Exam Signposting — Boundary Conditions

In exams, you may be asked to:

✓ State the boundary conditions at

- conductor–dielectric interface
- dielectric–dielectric interface

✓ Determine whether E_n or E_t changes across a boundary

✓ Identify where electric field is strongest in a multi-material system

You are expected to:

- Understand physical meaning
- Apply the conditions
- Not merely restate the postulates without interpretation

Designing a Cable: Breakdown & Safety Limits

Engineering question:
How thick must insulation be to prevent failure?

In this section, you will:
- Identify where the electric field is maximum in a cable
- Apply dielectric strength constraints in design
- Design insulation thickness based on voltage rating

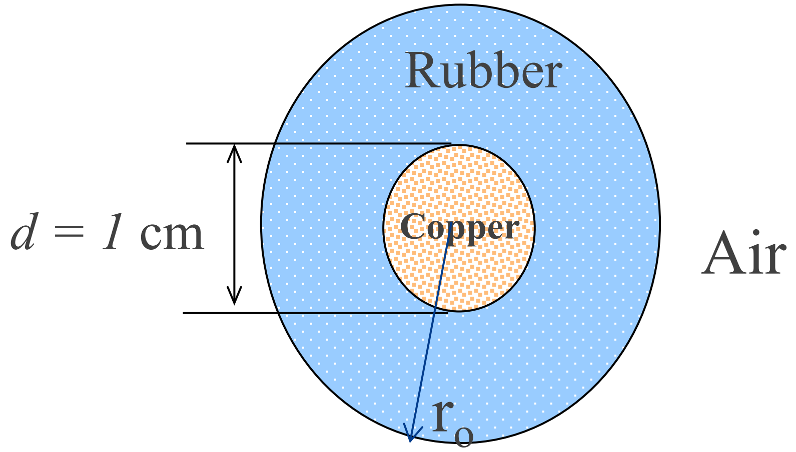
Example 5: Designing a high-voltage Cable

We have a copper conductor (diameter = 1 cm), rubber insulation ($\varepsilon_r = 3$), and air outside.

Breakdown strength: Rubber: 25 MV/m, Air: 3 MV/m

To ensure safety, we allow **only half** the breakdown strength during normal operation.

How to determine the thickness of the rubber insulation and maximum working voltage for the cable (voltage rating)

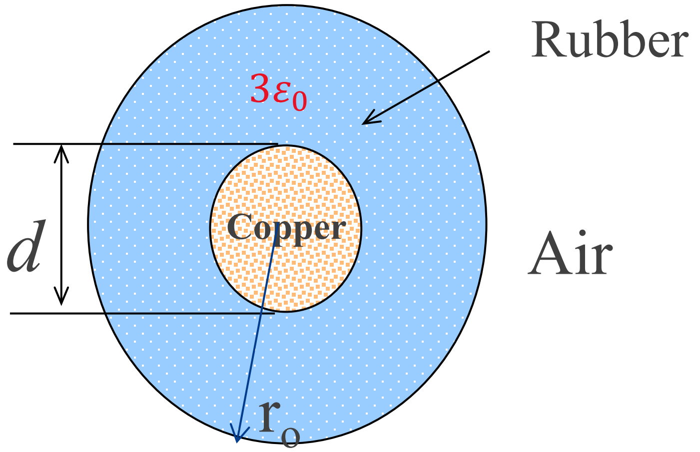


Where will the electric field be maximum? $|E| = \dfrac{\rho_l}{2\pi\varepsilon r}$

Thickness of the rubber = $r_o - d/2$

Step 1:Apply Maximum Field Constraint

$$|E| = \frac{\rho_l}{2\pi\varepsilon r}$$



At conductor surface $r = d/2 = 0.5$ cm

$$E_{\max\_r} = \frac{\rho_l}{2\pi\varepsilon \times d/2} = \frac{1}{2} \times 25 \text{ MV/m}$$

We can find the maximum allowable ρ_l from this condition

Step 2:Air breakdown Constraint Determines Outer Radius

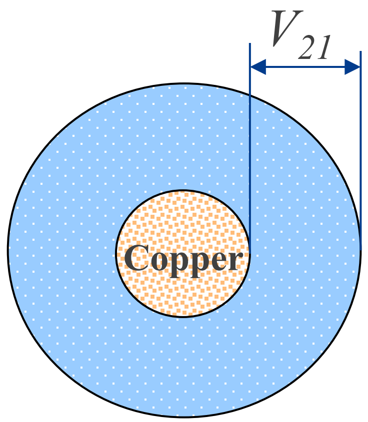
$$E_{\max\_a} = \frac{\rho_l}{2\pi\varepsilon_0 r_o} = \frac{1}{2} \times 3\text{MV/m}$$

This condition determines the required outer radius r_o

Required outer radius $r_o \approx 12.4$ cm
Required thickness $r_0 - d/2 = 11.9$ cm

Step 3: Determine potential

$$V_{21} = -\int_{r_1}^{r_2} \vec{E} \bullet d\vec{l} = \frac{\rho_l}{2\pi\varepsilon} \ln\left(\frac{r_1 = r_o}{r_2 = d/2}\right) = 200.7kV$$



Thus, the voltage rating of the cable is 200.7kV.

Required thickness = 11.9cmm
Rated voltage = 200.7kV

Why is the insulation thickness so large?

Cable design is governed by the weakest dielectric region.

⚠ Exam Signposting — Cable Design Problems
You may be required to:
✓ Identify where maximum electric field occurs
✓ Apply dielectric strength constraints
✓ Calculate required insulation thickness
✓ Determine maximum working voltage
Typical exam structure:
1. Locate maximum E
2. Apply breakdown limit
3. Solve for unknown geometry or charge
4. Compute voltage
Marks are awarded for:
- Correct modelling assumptions
- Clear reasoning
- Step-by-step logic

Pause & Reflect (Before we take a short break…)

✏️ **Take 1 minute to jot down:**

1. **One key idea** from this part of the lecture
2. **One question** that is still unclear
3. **Why is the insulation thickness determined by air region?**

💡 *This is for you — no submission.*

From Electric Field to Capacitance

We already know how to find:
•Electric field E
•Potential difference V

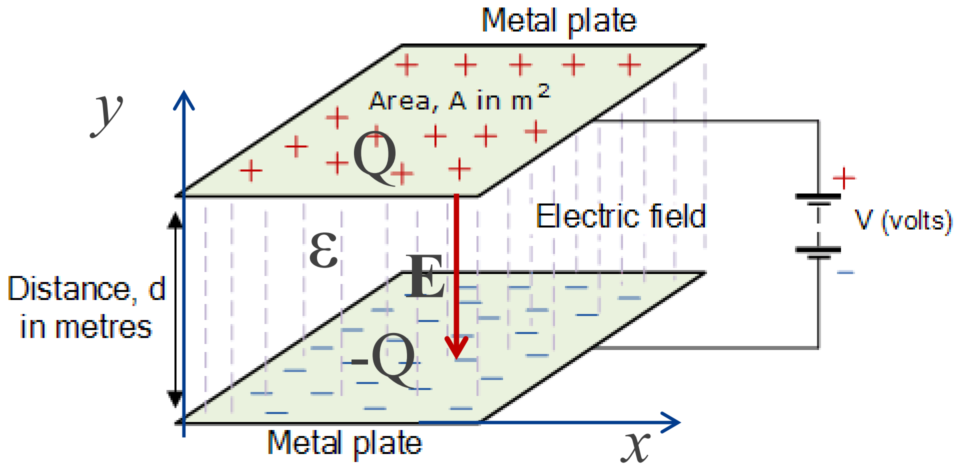
Now we ask:
How much charge is required to create that potential?

In this section, you will:
•Derive capacitance from field and voltage
•Analyse capacitance in transmission lines and coaxial cables
•Interpret capacitance per unit length

Capacitance formations

Let's us recall what we know about capacitance:

Recall- Parallel plate capacitor

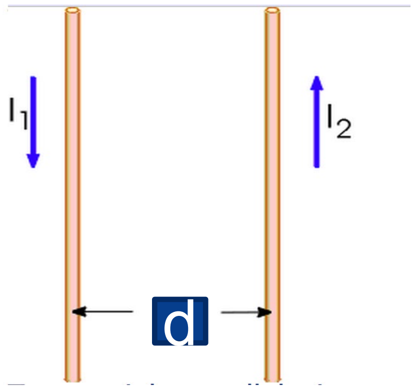


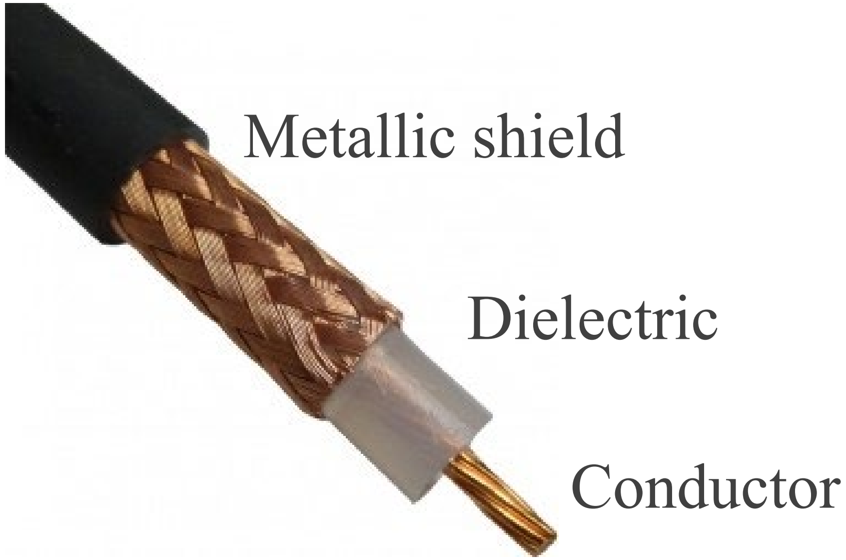
Find the electric field **E** between the plates with charge $\pm$Q:

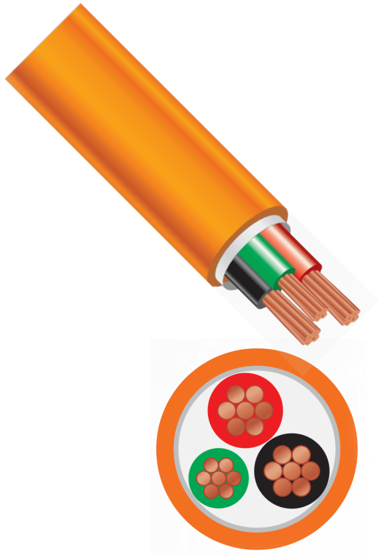
Find the potential form **E**

Find the capacitance

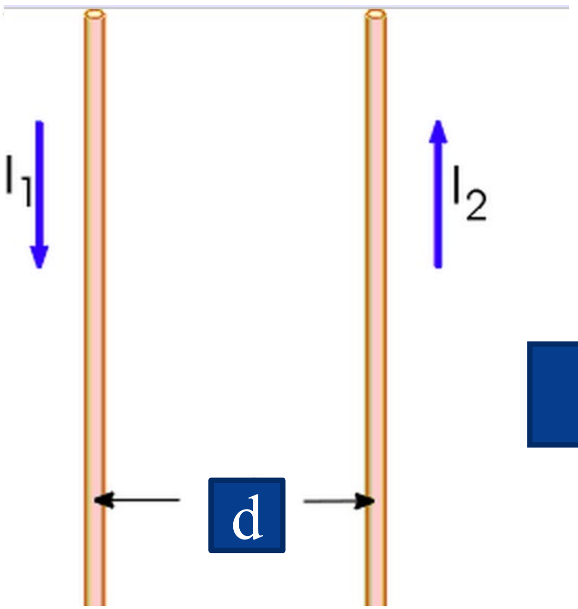
Can capacitance form in the following structures?

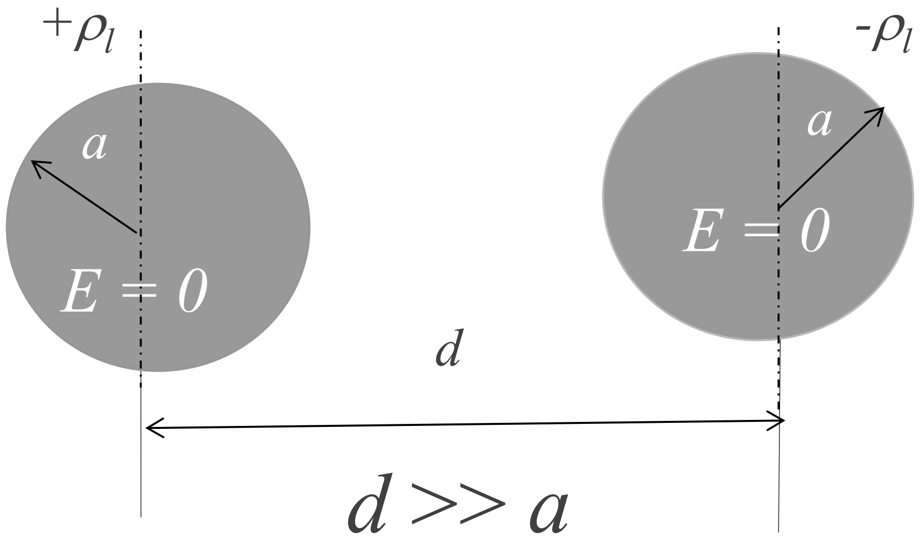






Case 1: Between two parallel conductors





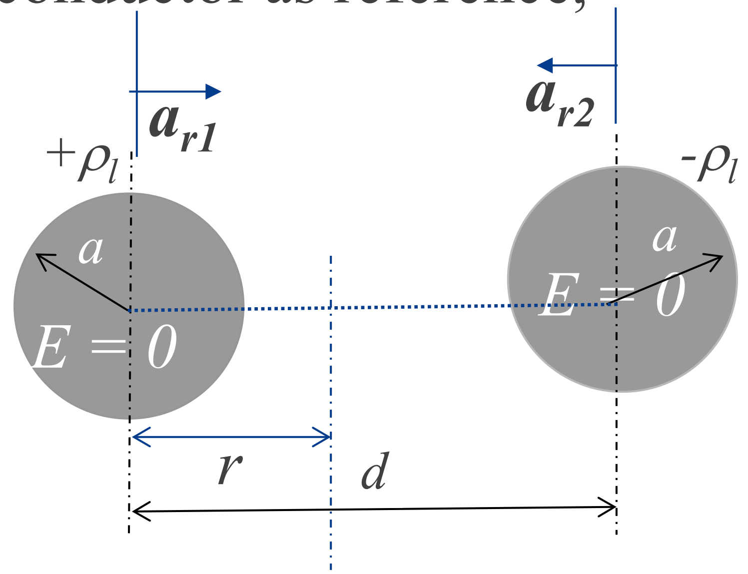
Consider the two conductors of radius a [m] to be very long and contain line charge density $\pm\rho_l$ [C/m].Centre to centre separation distance between two conductors is d [m].
If $d >> a$ and the conductors are uniformly charged, **E** at any radial distance r from a conductor can be found by Gauss's law.

$$\vec{E}_r = \vec{a}_r \frac{\pm\rho_l}{2\pi\varepsilon_0 r} \text{ V/m}$$

$$\vec{E}_{r1} = \vec{a}_{r1} \frac{\rho_l}{2\pi\varepsilon_0 r_1} \text{ V/m}, \quad \vec{E}_{r2} = \vec{a}_{r2} \frac{-\rho_l}{2\pi\varepsilon_0 r_2} \text{ V/m}$$

where, r_1 and r_2 are arbitrary radial distances defined from centres of conductor 1 and 2 respectively.

Potential *at conductor 1* due to $+\rho_l$ considering other conductor as reference,



Note the direction of unit vectors $\boldsymbol{a}_{r1}$ and $\boldsymbol{a}_{r2}$

$$V_1 = -\int_{d-a_2}^{a_1} \vec{E}_{r1} \bullet d\vec{r} = \left[\frac{\rho_l}{2\pi\varepsilon_0} \ln \frac{d-a}{a} \right] \text{ Volt}$$

Note: conductor radius $a_1 = a_2 = a$

Similarly, potential at the conductor 2 due to $-\rho_l$,

$$V_2 = -\int_{d-a_1}^{a_2} \vec{E}_{r2} \bullet d\vec{r} = \left[\frac{-\rho_l}{2\pi\varepsilon_0} \ln \frac{d-a}{a} \right] \text{ Volt}$$

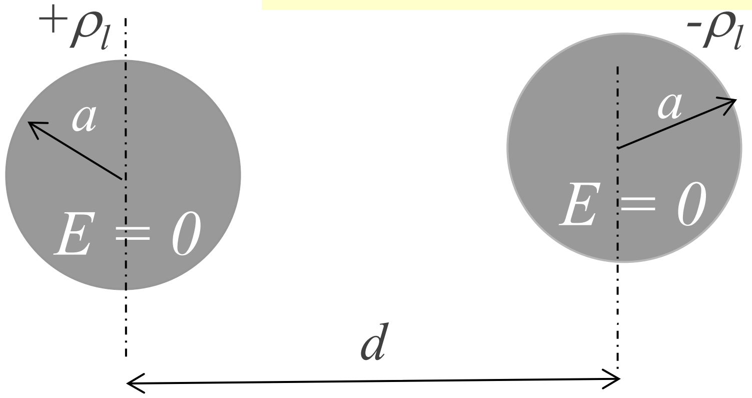
The potential difference between the two conductors :

$$\therefore V_{12} = V_1 - V_2 = \left(\frac{\rho_l}{\pi\varepsilon_0} \ln \frac{d-a}{a} \right) \text{ Volt}$$

Capacitance between the conductors $C = \dfrac{Q}{V}$ [F]

$$V_{12} = \left(\frac{\rho_l}{\pi\varepsilon_0} \ln \frac{d-a}{a} \right) \text{ Volt}$$

For long transmission lines capacitance C has a very large value.



Capacitance per meter [F/m] is used more often for such long lines.

$$\therefore C = \frac{\rho_l}{V_{12}} = \frac{\pi\varepsilon_0}{\ln \dfrac{d-a}{a}} \quad \text{F/m}, \quad \text{If } d >> a, \quad C \approx \frac{\pi\varepsilon_0}{\ln \dfrac{d}{a}} \text{ F/m}$$

If distance d increases, what happens to capacitance?

⚠ Exam Signposting — Capacitance Derivations

You may be asked to:

✓ Derive capacitance from electric field and potential

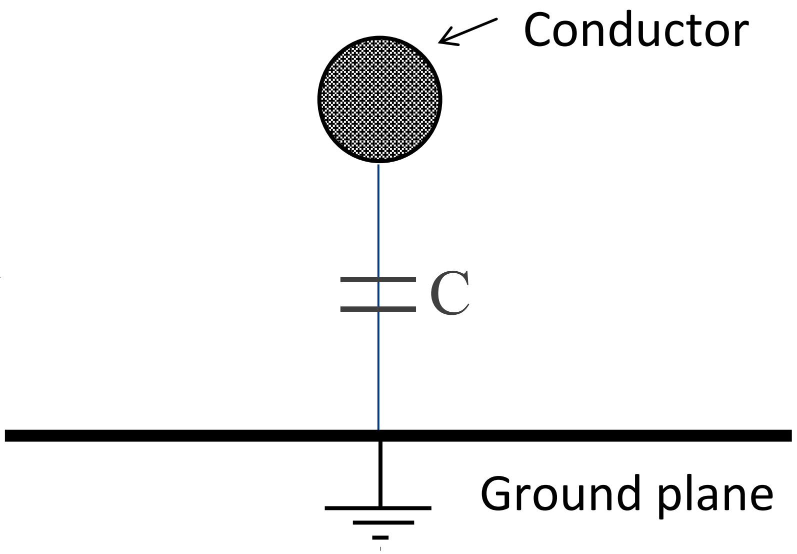
✓ Compute capacitance per unit length

✓ Compare capacitance for different geometries

You are expected to: Start from known $E(r)$, Integrate to find V, Use $C = Q/V$

Not required: ✕ Memorising final expressions without derivation

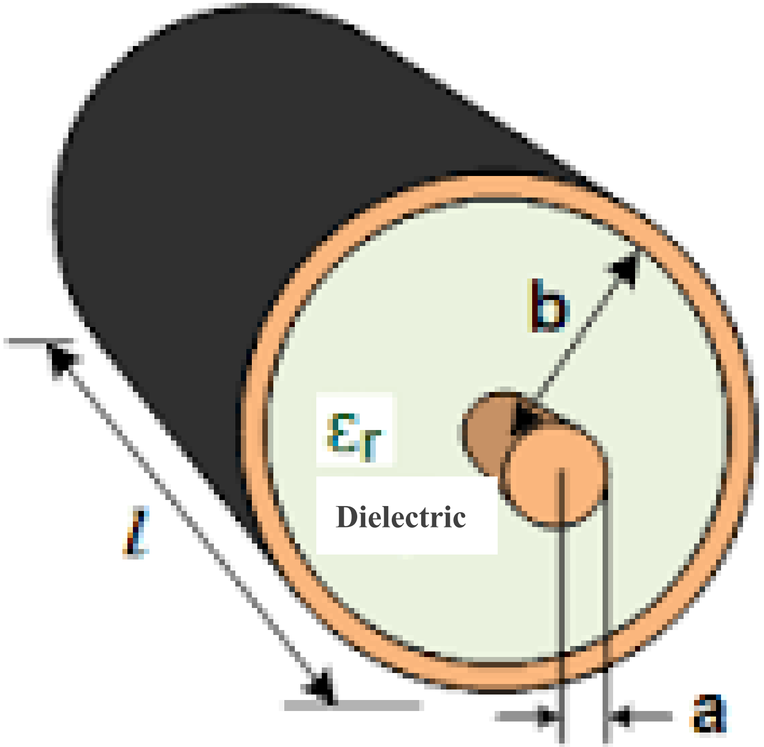
Case 2: Between a conductor and a ground plane



Recall example 2 when finding E due the charged conductor using Gauss's law, we assumed there is no effect of the ground plane. Now if we want to investigate capacitance, we cannot neglect the effect of the ground. Finding of **E**, V and C is more complex. We will deal with this problem while discussing '*Method of image*' in Topic 2.

We cannot neglect ground when calculating capacitance — unlike when calculating field far away from it.

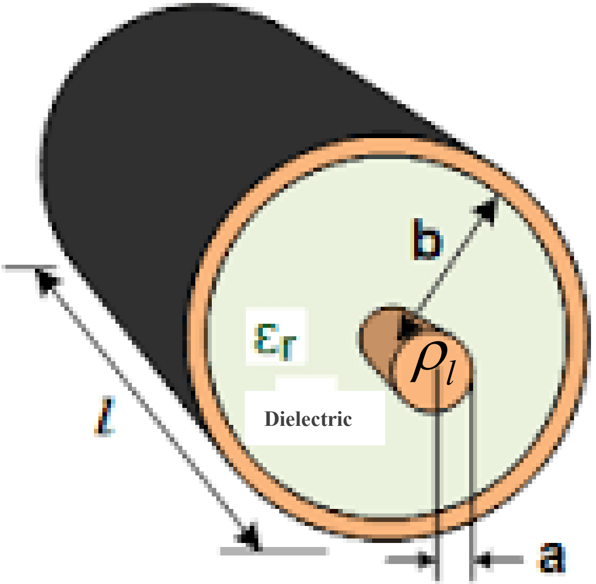
Case 3: Between the conductor and the metallic shield



Consider the co-axial cable with a conductor of radius a has charge density ρ_l C/m. The radius to the inner side of the metallic shield is b and the relative permittivity of the dielectric material is ε_r.

1. E at any redial distance r from the surface of the conductor within the dielectric i.e. $a < r < b$:

$$\vec{E} = \vec{a}_r \frac{\rho_l}{2\pi\varepsilon r} \text{ V/m, where } a < r < b$$



2. Potential V between the conductor and the shield,

$$V_{ab} = -\int_b^a \vec{E} \bullet d\vec{r} = -\int_b^a \left(\vec{a}_r \frac{\rho_l}{2\pi\varepsilon r} \right) \bullet \left(\vec{a}_r dr \right) = \frac{\rho_l}{2\pi\varepsilon} \ln\left(\frac{b}{a} \right)$$

3. Capacitance per unit length:

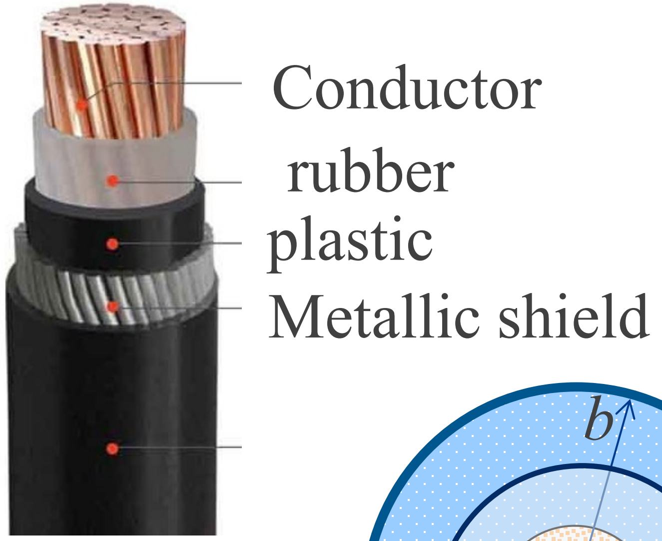
$$C = \frac{\rho_l}{V_{ab}} = \frac{2\pi\varepsilon}{ln\left(\frac{b}{a}\right)} \text{ F/m}$$

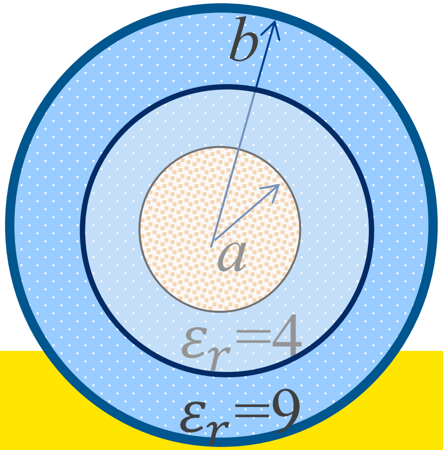
⚠ **Exam Signposting — Coaxial Cable**
For coaxial systems, be able to:
✓ Identify the region where field exists (a < r < b)
✓ Apply Gauss' law
✓ Derive potential difference
✓ Compute capacitance per unit length
Common mistake to avoid:
•Forgetting limits of integration
•Confusing conductor radius with shield radius

Example 6

A long co-axial cable is made with a conductor of radius a = 2 mm and shield internal radius b = 5 mm. There are two layers of insulation between the conductor and the shield. The first insulation layer is 1mm thick and made of rubber (ε_r= 4). The second layer is plastic (ε_r=9) and 2 mm thick. Calculate the capacitance per unit length of the cable.

Ans: 352pF/m





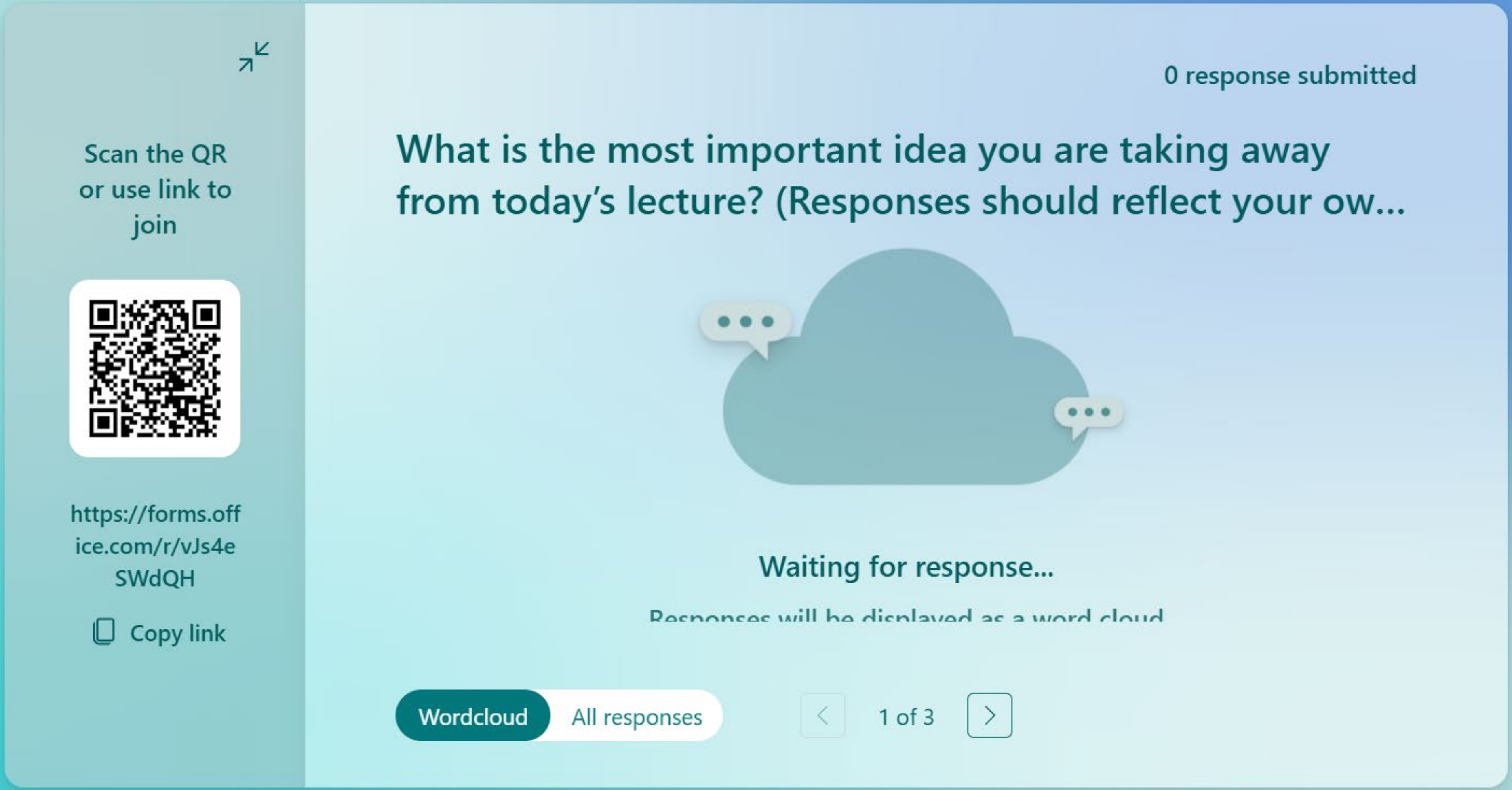
1. Field $$\vec{E}_r = \vec{a}_r \frac{\rho_l}{2\pi\varepsilon r} \text{ V/m}, \ \varepsilon = \varepsilon_0\varepsilon_r, \varepsilon_0 = 8.854\times10^{-12}\text{ F/m}$$

2. Potential $$V_{ab} = -\int_{b}^{b-thickness} \vec{E}_{plastic} \bullet d\vec{r} - \int_{b-thickness}^{a} \vec{E}_{rubber} \bullet d\vec{r}$$

2. capacitance $$C = \frac{\rho_l}{V_{ab}}$$

Before You Go… (1 minute reflection)

This reflection will count toward optional bonus marks.



Today’s Big Takeaways

1. Boundary conditions determine field behaviour at interfaces.

2. Breakdown strength constrains design (specifically cable design).

3. Capacitance depends on geometry and permittivity.

4. Capacitance can form between transmission lines and in coaxial cables.

Review and further practice (Recommended)

Practice Examples 3-14, 3-15, 3-16 from the textbook by D.K. Cheng.

Attempt as a challenge:

Problems P.3-26, P.3-28, P.3-30, P.3-35, P.3-36, P.3-37 from the textbook by D.K. Cheng.

🎯 **What You Must Be Able To Do After Today**

1. Apply boundary conditions correctly.
2. Analyse electric field behaviour in multi-layer cables.
3. Use dielectric strength to design safe insulation thickness.
4. Derive capacitance from field and voltage.
5. Interpret capacitance per unit length in transmission lines.

If you can do these five things, you are exam-ready for this topic.